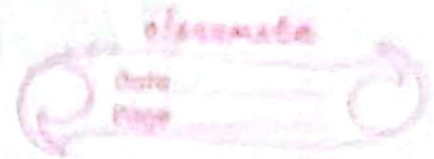


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Calculate y'

1) $y = (x^2 + x^3)^4$
Soln:-

$$y' = \frac{dy}{dx} = \frac{d(x^2 + x^3)^4}{d(x^2 + x^3)} \cdot \frac{d(x^2 + x^3)}{dx}$$

$$\begin{aligned} &= 4(x^2 + x^3)^3 \cdot (2x + 3x^2) \\ &= 4(x^2(1 + x^2))^3 (x(3x + 2)) \\ &= 4x^6(1 + x^2)^3 \cdot x(3x + 2) \\ &= 4x^7(1 + x^2)^3(3x + 2) // \end{aligned}$$

$$\therefore y' = 4x^7(1 + x^2)^3(3x + 2)$$

$$b) y = \frac{1}{\sqrt{x}} - \frac{1}{\sqrt[5]{x^3}}$$

Soln:-

$$y = \frac{1}{\sqrt{x}} - \frac{1}{\sqrt[5]{x^3}}$$

$$y = \frac{x^0}{\sqrt{x}} - \frac{x^0}{\sqrt[5]{x^3}}$$

$$y = \frac{x^0}{x^{1/2}} - \frac{x^0}{x^{3/5}}$$

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$$Q) y = x^{0 - \frac{1}{2}} - x^{0 - \frac{3}{5}}$$

$$Q) y = x^{-\frac{1}{2}} - x^{-\frac{3}{5}}$$

Differentiating y w.r.t. x we get,

$$Q) \frac{dy}{dx} = \frac{dx^{-\frac{1}{2}}}{dx} - \frac{dx^{-\frac{3}{5}}}{dx}$$

$$Q) \frac{dy}{dx} = \left(-\frac{1}{2}\right)x^{-\frac{1}{2}-1} - \left(-\frac{3}{5}\right)x^{-\frac{3}{5}-1}$$

$$Q) \frac{dy}{dx} = \left(-\frac{1}{2}\right)x^{-\frac{1}{2}-2} + \frac{3}{5}x^{-\frac{3}{5}-5}$$

$$Q) \frac{dy}{dx} = \left(-\frac{1}{2}\right)x^{-\frac{5}{2}} + \frac{3}{5}x^{-\frac{8}{5}}$$

$$Q) \frac{3}{5}x^{-\frac{8}{5}} - \frac{1}{2}x^{-\frac{5}{2}} = \frac{dy}{dx}$$

$$\therefore y' = \frac{dy}{dx} = \frac{3}{5}x^{-\frac{8}{5}} - \frac{1}{2}x^{-\frac{5}{2}}$$

3) $y = \frac{x^2 - x + 2}{\sqrt{x}}$

Soln:-

$$y = \frac{x^2 - x + 2}{\sqrt{x}}$$

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$$\text{cn } y = \frac{x^2}{x^{1/2}} - \frac{x}{x^{1/2}} + \frac{2}{x^{1/2}}$$

$$\text{cn } y = x^{\frac{2-1}{2}} - x^{\frac{1-1}{2}} + 2x^{\frac{-1}{2}}$$

$$\text{cn } y = x^{\frac{4-1}{2}} - x^{\frac{2-1}{2}} + 2x^{\frac{-1}{2}}$$

$$\text{cn } y = x^{\frac{3}{2}} - x^{\frac{1}{2}} + 2x^{\frac{-1}{2}}$$

Differentiating y w.r.t. x we get,

$$\text{cn } \frac{dy}{dx} = \frac{d}{dx} x^{\frac{3}{2}} - \frac{d}{dx} x^{\frac{1}{2}} + 2 \frac{d}{dx} x^{\frac{-1}{2}}$$

$$\text{cn } \frac{dy}{dx} = \frac{3}{2} x^{\frac{3-1}{2}} - \frac{1}{2} x^{\frac{1-1}{2}} + 2 \cdot \left(\frac{-1}{2}\right) x^{\frac{-1-1}{2}}$$

$$\text{cn } \frac{dy}{dx} = \frac{3}{2} x^{\frac{3-2}{2}} - \frac{1}{2} x^{\frac{1-2}{2}} - x^{\frac{-1-2}{2}}$$

$$\text{cn } \frac{dy}{dx} = \frac{3}{2} x^{\frac{1}{2}} - \frac{1}{2} x^{\frac{-1}{2}} - x^{\frac{-3}{2}}$$

$$\text{cn } \frac{dy}{dx} = 3\sqrt{x} - \frac{1}{2\sqrt{x}} - \frac{1}{x^{3/2}}$$

4) $y = \frac{\tan n}{1 + \cos n}$

Soln:-

Differentiating y w.r.t. n we get,

$$\text{or } \frac{dy}{dn} = \frac{1 + \cos n \frac{d \tan n}{dn} - \tan n \frac{d(1 + \cos n)}{dn}}{(1 + \cos n)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{\sec^2 n (1 + \cos n) - \tan n (0 - \sin n)}{(1 + \cos n)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{\sec^2 n (1 + \cos n) + \sin n \tan n}{(1 + \cos n)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{\sec^2 n (1 + \cos n)}{(1 + \cos n)^2} + \frac{\sin n \tan n}{(1 + \cos n)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{\sec^2 n}{1 + \cos n} + \frac{\sin n \tan n}{(1 + \cos n)^2}$$

5) $y = n^2 \sin \pi n$

Soln:-

Differentiating y w.r.t. n we get,

$$\text{or } \frac{dy}{dn} = n^2 \frac{d \sin \pi n}{d(\pi n)} \frac{d(\pi n)}{dn} + \sin(\pi n) \frac{dn^2}{dn}$$

$$\text{or } \frac{dy}{dn} = \pi n^2 \cos \pi n + 2n \sin(\pi n)$$

$$\text{on } \frac{dy}{dx} = x(\pi \cos \pi x + 2 \sin \pi x)$$

6) $y = x \cos^{-1} x$
Soln:-

$$y = x \cos^{-1} x$$

Differentiating y w.r.t. x we get,

$$\frac{dy}{dx} = x \frac{d(\cos^{-1} x)}{dx} + \cos^{-1} x \frac{dx}{dx}$$

$$\frac{dy}{dx} = x \left(\frac{-1}{\sqrt{1-x^2}} \right) + \cos^{-1} x$$

$$\frac{dy}{dx} = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

7)

$$y = \frac{t^4 - 1}{t^4 + 1}$$

Soln:-

$$y = \frac{t^4 - 1}{t^4 + 1}$$

Diff. y w.r.t. t we get,

$$\frac{dy}{dt} = \frac{t^4 + 1 \frac{d(t^4 - 1)}{dt} - (t^4 - 1) \frac{d(t^4 + 1)}{dt}}{(t^4 + 1)^2}$$

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$$\frac{dy}{dt} = \frac{t^4 + 1 [4t^3] - (4t^3)(t^4 - 1)}{(t^4 + 1)^2}$$

$$\frac{dy}{dt} = \frac{4t^7 + 4t^3 - 4t^7 + 4t^3}{(t^4 + 1)^2}$$

$$\frac{dy}{dt} = \frac{8t^3}{(t^4 + 1)^2}$$

$$\therefore y' = \frac{8t^3}{(t^4 + 1)^2}$$

8) $ne^y = y \sin n$
Soln:-

Differentiating both sides w.r.t. n we get,

$$\frac{d(ne^y)}{dn} = \frac{d(y \sin n)}{dn}$$

$$n \frac{de^y}{dy} \cdot \frac{dy}{dn} + e^y \frac{dn}{dn} = y \frac{d \sin n}{dn} + \sin n \frac{dy}{dn}$$

$$n e^y \frac{dy}{dn} + \sin n = y \cos n + \sin n \frac{dy}{dn}$$

$$\frac{dy}{dn} (ne^y - \sin n) = y \cos n - e^y$$

$$\frac{dy}{dn} = \frac{y \cos n - e^y}{ne^y - \sin n}$$

$$\therefore y' = \frac{y \cos n - e^y}{ne^y - \sin n}$$

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9) $y = \ln(x \ln x)$
Soln:-

$$y = \ln(x \ln x)$$

Diff. y w.r.t. x we get

$$\frac{dy}{dx} = \frac{d \ln(x \ln x)}{d(x \ln x)} \cdot \frac{d(x \ln x)}{dx}$$

$$\text{or } \frac{dy}{dx} = \frac{1}{x \ln x} \left[x \frac{d \ln x}{dx} + \ln x \frac{dx}{dx} \right]$$

$$\text{or } \frac{dy}{dx} = \frac{1}{x \ln x} \left[x \cdot \frac{1}{x} + \ln x \cdot 1 \right]$$

$$\text{or } \frac{dy}{dx} = \frac{1 + \ln x}{x \ln x}$$

$$\therefore y' = \frac{1 + \ln x}{x \ln x}$$

10) $y = e^{mx} \cos nx$
Soln:-

$$y = e^{mx} \cos nx$$

Diff. y w.r.t. x we get,

$$\frac{dy}{dx} = e^{mx} \frac{d(\cos nx)}{dx} + \cos nx \cdot \frac{d e^{mx}}{dx}$$

$$\frac{dy}{dx} = -n e^{mx} \sin nx + m \cos nx e^{mx}$$

$$\therefore y' = m \cos nx e^{mx} - n e^{mx} \sin nx$$

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61-63) Find an equation of the tangent line to the curve at the given point.

61) $y = 4 \sin^2 x$, $(\pi/6, 1)$
Soln:-

The given equation is:

$$y = 4 \sin^2 x \quad \dots (i)$$

Diff. eqn (i) w.r.t. x we get,

$$\frac{dy}{dx} = 4 \frac{d \sin^2 x}{dx} = 8 \sin x \cos x$$

$$\frac{dy}{dx} = 4 \cos 2x$$

$$\frac{dy}{dx} = 4 \times 2 \sin x \cdot \cos x$$

$$\frac{dy}{dx} = 8 \sin x \cos x$$

Put $x = \frac{\pi}{6}$

$$\frac{dy}{dx} = 8 \sin\left(\frac{\pi}{6}\right) \cos\left(\frac{\pi}{6}\right)$$

$$\frac{dy}{dx} = 8 \times \frac{1}{2} \times \frac{\sqrt{3}}{2}$$

$$\frac{dy}{dx} = 2\sqrt{3}$$

$$\therefore m = 2\sqrt{3}$$

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Using equation of normal form

$$y - y_1 = m(x - x_1)$$

$$\text{cn } y - 1 = 2\sqrt{3}\left(x - \frac{\pi}{6}\right)$$

$$\text{cn } y - 1 = 2\sqrt{3}x - \frac{2\sqrt{3}\pi}{6}$$

$$\text{cn } y - 1 = 2\sqrt{3}x - \frac{\sqrt{3}\pi}{3}$$

$$\text{cn } y - 1 = 2\sqrt{3}x - \frac{\sqrt{3}\pi}{\sqrt{3} \cdot \sqrt{3}}$$

$$\text{cn } \sqrt{3}(y - 1) = (2\sqrt{3} \cdot \sqrt{3})x - \pi$$

$$\text{cn } \sqrt{3}y - \sqrt{3} = 2 \times 3x - \pi$$

$$\text{cn } 6x - \sqrt{3}y - \pi + \sqrt{3} = 0$$

$$\text{Q 2) } y = \frac{x^2 - 1}{x^2 + 1}, (0, -1)$$

Soln:-

$$\frac{dy}{dx} = \frac{(x^2 + 1) \frac{d(x^2 - 1)}{dx} - (x^2 - 1) \frac{d(x^2 + 1)}{dx}}{(x^2 + 1)^2}$$

$$\text{cn } \frac{dy}{dx} = \frac{2x(x^2 + 1) - [2x(x^2 - 1)]}{(x^2 + 1)^2}$$

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$$\text{or } \frac{dy}{dn} = \frac{2n^3 + 2n - [2n^3 - 2]}{(n^2 + 1)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{2n^3 + 2n - 2n^3 + 2}{(n^2 + 1)^2} = \frac{2n + 2}{(n^2 + 1)^2}$$

$$\text{or } \frac{dy}{dn} = \frac{4n}{(n^2 + 1)^2}$$

$$\text{Put } n = 0$$

$$\frac{dy}{dn} = \frac{4 \times 0}{0 + 1} = 0$$

$$\therefore \text{Slope}(m) = 0$$

Using point-slope form,

$$y - y_1 = m(n - n_1)$$

$$y - (-1) = 0$$

$$y = -1$$

83) $y = \sqrt{1 + 4 \sin n}$, $(0, 1)$
Soln:-

$$y = \sqrt{1 + 4 \sin n}$$

$$\frac{dy}{dn} = \frac{d(1 + 4 \sin n)^{\frac{1}{2}}}{d(1 + 4 \sin n)} = \frac{d(1 + 4 \sin n)}{d(1 + 4 \sin n)}$$

$$\frac{dy}{dn} = \frac{1}{2} (1 + 4 \sin n)^{\frac{1}{2} - 1} \cdot [0 + 4 \cos n]$$

$$\frac{dy}{dn} = \frac{1}{2} \cdot \frac{4 \cos n}{\sqrt{1 + 4 \sin n}}$$

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$$\frac{dy}{dx} = \frac{2 \cos n}{\sqrt{1+4 \sin n}}$$

Put $n = 0$

$$\frac{dy}{dx} = \frac{2 \cos 0}{\sqrt{1+4 \sin 0}}$$

$$\frac{dy}{dx} = \frac{2 \times 1}{\sqrt{1+4 \times 0}}$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1}} = 2$$

Now

Using normal intercept formulae,

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 0)$$

$$y - 1 = 2x$$

$$2x - y + 1 = 0$$

64-65) Find the equations of the tangent line and normal line to the curve at the given points

64) $x^2 + 4xy + y^2 = 13, (2, 1)$

Soln:-

The given equation is

$$x^2 + 4xy + y^2 = 13$$

Diff both sides w.r.t. x we get

$$\frac{d(x^2 + 4xy + y^2)}{dx} = \frac{d(13)}{dx}$$

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$$m \quad \frac{dx^2}{dx} + \frac{d(4xy)}{dx} + \frac{dy^2}{dy} \cdot \frac{dy}{dx} = 0$$

$$m \quad 2x + 4 \left[x \frac{dy}{dx} + y \frac{dx}{dx} \right] + 2y \cdot \frac{dy}{dx} = 0$$

$$m \quad 2x + 4x \frac{dy}{dx} + 4y + 2y \frac{dy}{dx} = 0$$

$$m \quad 2x + 4y = -2y \frac{dy}{dx} - 4x \frac{dy}{dx}$$

$$m \quad 2x + 4y = -2 \frac{dy}{dx} [y + 2x]$$

$$m \quad x(y + 2x) = - \frac{dy}{dx} (y + 2x) \cdot y - y$$

$$m \quad \frac{x + 2y}{y + 2x} = \frac{dy}{dx} \cdot \frac{y + 2x}{y + 2x}$$

$$m \quad \frac{-x - 2y}{y + 2x} = \frac{dy}{dx} \cdot \frac{y + 2x}{y + 2x}$$

$$\text{Put } x = 2x, y = 1$$

$$\frac{dy}{dx} = \frac{-2 - 2x}{1 + 2x}$$

$$\frac{dy}{dx} = \frac{-2 - 2}{1 + 4} = -\frac{4}{5}$$

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$$\therefore \text{Slope}(m) = -\frac{4}{5}$$

Equation of a tangent is

$$y - y_1 = m(x - x_1)$$
$$y - 1 = -\frac{4}{5}(x - 2)$$

$$5(y - 1) = -4x + 8$$

$$5y - 5 = -4x + 8$$

$$4x + 5y - 5 - 8 = 0$$

$$4x + 5y - 13 = 0 //$$

Equation of a normal is given by

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$y - 1 = -\frac{1}{-\frac{4}{5}}(x - 2)$$

$$y - 1 = \frac{5}{4}(x - 2)$$

$$4y - 4 = 5x - 10$$

$$5x - 4y - 10 + 4 = 0$$

$$5x - 4y - 6 = 0 //$$

$\therefore$ The equation of tangent is $4x + 5y - 13 = 0$
and normal is $5x - 4y - 6 = 0$

65) $y = (2+x)e^{-x}$, (0, 2)
Soln:-

The given equation is $y = (2+x)e^{-x}$.
Diff. eqⁿ (i) w.r.t. x we get,

$$\frac{dy}{dx} = \frac{d(2+x)}{dx} \cdot e^{-x} + (2+x) \frac{d(e^{-x})}{d(-x)} \cdot \frac{d(-x)}{dx}$$

$$= [0+1] \cdot e^{-x} + (-1) \cdot e^{-x} (2+x)$$

$$= e^{-x} + (2+x)(-e^{-x})$$

$$= e^{-x} - xe^{-x} - 2e^{-x}$$

$$= -e^{-x} - xe^{-x}$$

Put $x = 0$

$$\frac{dy}{dx} = -e^0 - xe^0 = -1 - x$$

Put $x = 0$

$$\frac{dy}{dx} = -1 = m \text{ (slope)}$$

Using tangent equation form

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -1(x - 0)$$

$$y - 2 = -x$$

$$y + x - 2 = 0$$

∴ The equation of tangent is $x + y - 2 = 0$

The equation of normal is

$$y - 2 = \frac{-1}{-1}(x - 0)$$

$$y - 2 = x \quad x - y + 2 = 0$$

The equation of normal is $x - y + 2 = 0$

60) If $f(x) = xe^{\sin x}$, find $f(x)$ Graph f and f' on the same screen and comment

Soln:-

The function is $f(x) = xe^{\sin x}$

$$y = xe^{\sin x} \quad \text{--- (i)}$$

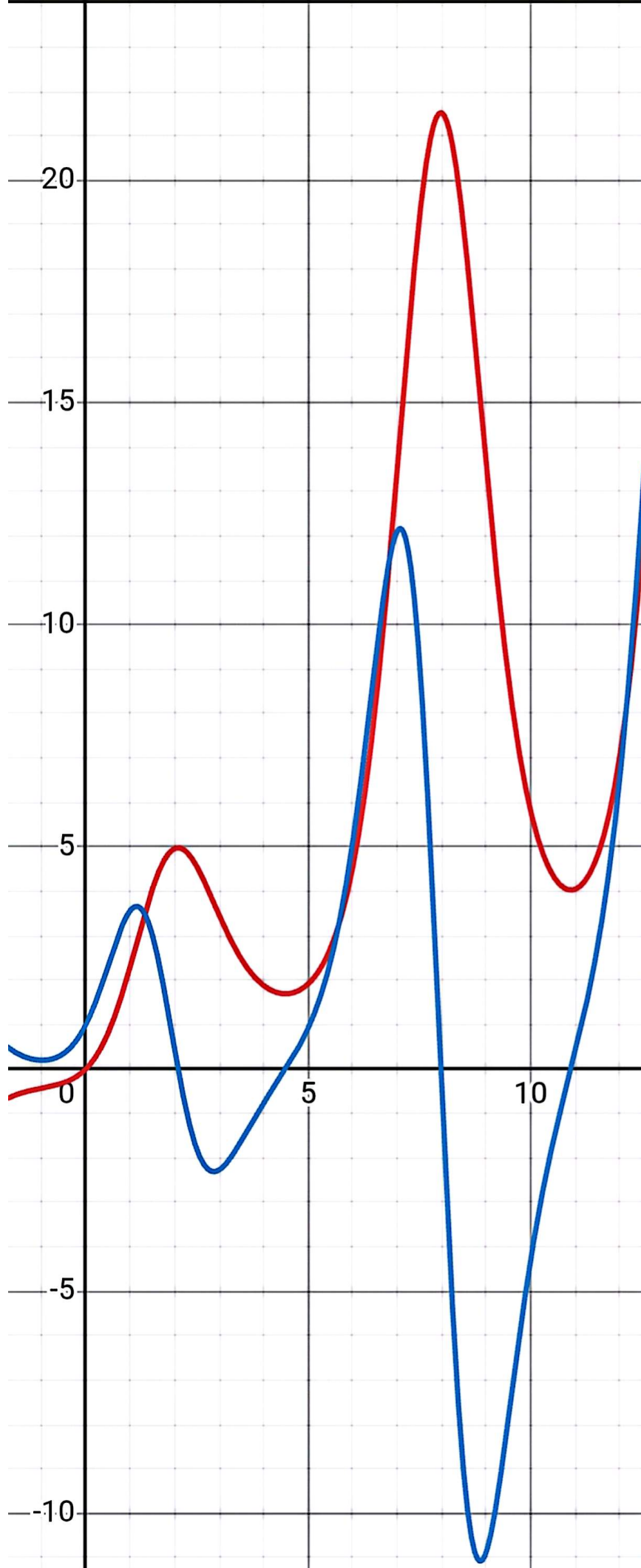
Differentiating eqn (i) w.r.t. x we get,

$$\frac{dy}{dx} = \frac{d(xe^{\sin x})}{dx}$$

$$= x \frac{de^{\sin x}}{d\sin x} \cdot \frac{d\sin x}{dx} + e^{\sin x} \frac{dx}{dx}$$

$$= x \cos x e^{\sin x} + e^{\sin x}$$

$$= e^{\sin x} (x \cos x + 1)$$



The red graph is the function $f(x) = x e^{\sin x}$ and the blue graph is its derivative $f'(x) = e^{\sin x} (x \cos x + 1)$. When the blue graph is above the x -axis ($f'(x) > 0$), the red line is increasing and when the blue graph is below the x -axis ($f'(x) < 0$), the red graph is decreasing. The points where the blue graph cuts the x -axis ($f'(x) = 0$) give the critical points of the red graph. If the blue graph changes from positive to negative, the red graph has a local maximum and if it changes from negative to positive, the red graph has a local minimum.

67) If $f(x) = x\sqrt{5-x}$, find $f'(x)$

Soln:-

$$f(x) = x\sqrt{5-x}$$

$$y = x\sqrt{5-x}$$

Differentiating y w.r.t. x we get,

$$\frac{dy}{dx} = x \frac{d(\sqrt{5-x})}{dx} + \sqrt{5-x} \frac{dx}{dx}$$

$$\frac{dy}{dx} = x \cdot \left(-\frac{1}{2}\right) (5-x)^{-\frac{1}{2}} (0-1) + \sqrt{5-x}$$

$$\frac{dy}{dx} = -\frac{x}{2\sqrt{5-x}} + \sqrt{5-x}$$

$$\frac{dy}{dx} = \frac{-x + 2(5-x)}{2\sqrt{5-x}}$$

$$\frac{dy}{dx} = \frac{-x + 10 - 2x}{2\sqrt{5-x}}$$

$$\frac{dy}{dx} = \frac{-3x + 10}{2\sqrt{5-x}}$$

b) Find equation of the tangent line to the curve $y = x\sqrt{5-x}$ at points $(1, 2)$ and $(4, 4)$

Soln:-

The given equation is $y = \frac{10-3x}{2\sqrt{5-x}}$

at point $(1, 2)$

$$m = \frac{10 - 3x}{2\sqrt{5-x}} = \frac{7}{4}$$

Equation of tangent line at (1, 2)

$$y - 2 = \frac{7}{4}(x - 1)$$

$$4y - 8 = 7x - 7$$

$$7x - 4y - 7 + 8 = 0$$

$$7x - 4y + 1 = 0$$

at point (4, 4)

$$m = \frac{10 - 3x}{2\sqrt{5-x}} = \frac{-2}{2} = -1$$

Equation of tangent line at 4, 4

$$y - y_1 = m(x - x_1)$$

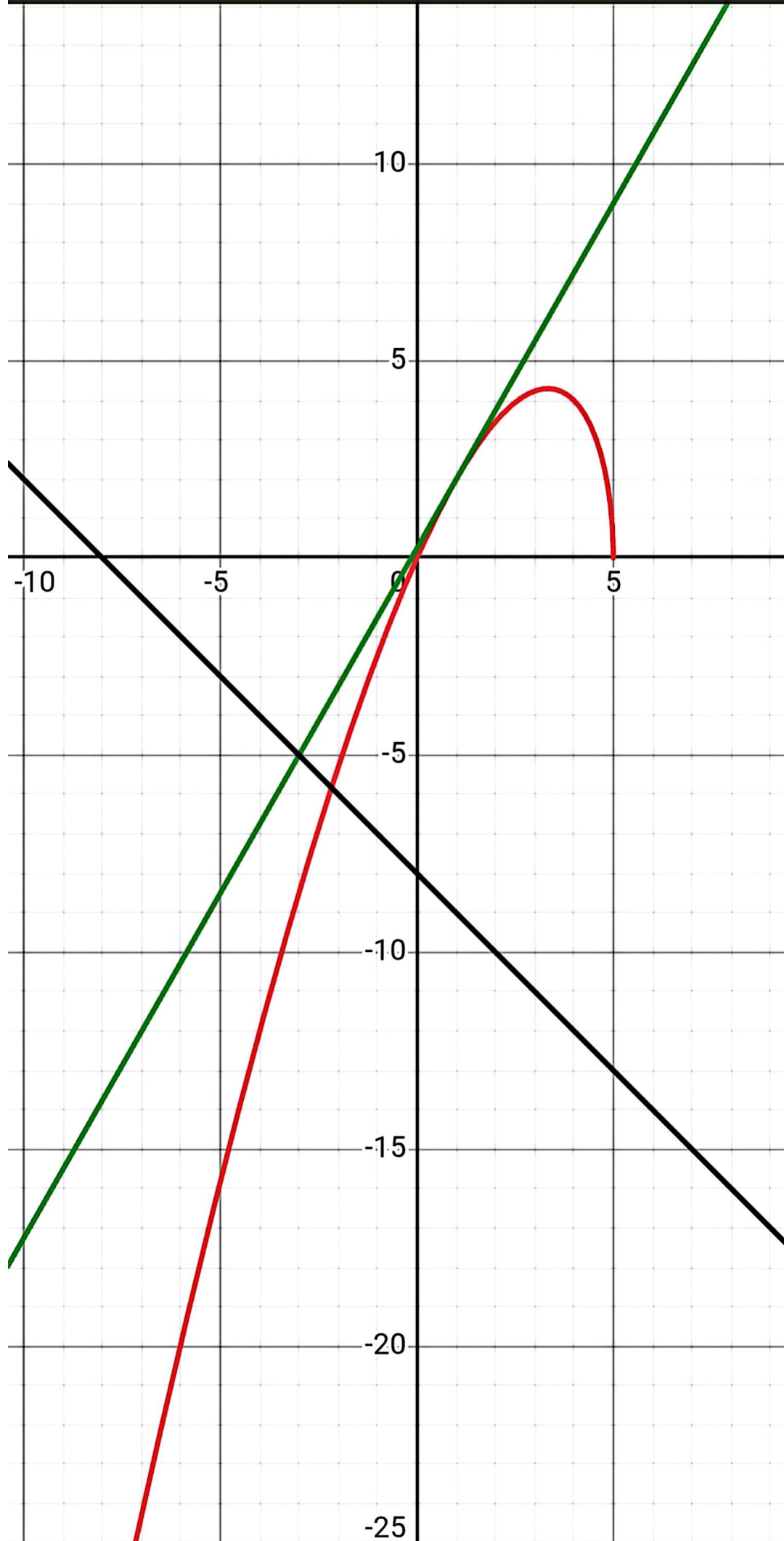
$$y - 4 = -1(x - 4)$$

$$y - 4 = -x + 4$$

$$y + x - 8 = 0$$

c) Illustrate part (b) by graphing the curve and tangent lines on the same screen.

d) check to see that your answer to part (a) is reasonable by comparing the graphs of f and f' .





6x)a) If $f(x) = 4x - \tan x$ $-\pi/2 < x < \pi/2$, find f' and f''

Solⁿ -

$$y = 4x - \tan x \quad \dots (i)$$

Differentiating eqn (i) w.r.t. x we get,

$$\frac{dy}{dx} = 4 - \sec^2 x$$

or

$$y' = 4 - \sec^2 x \quad \dots (ii)$$

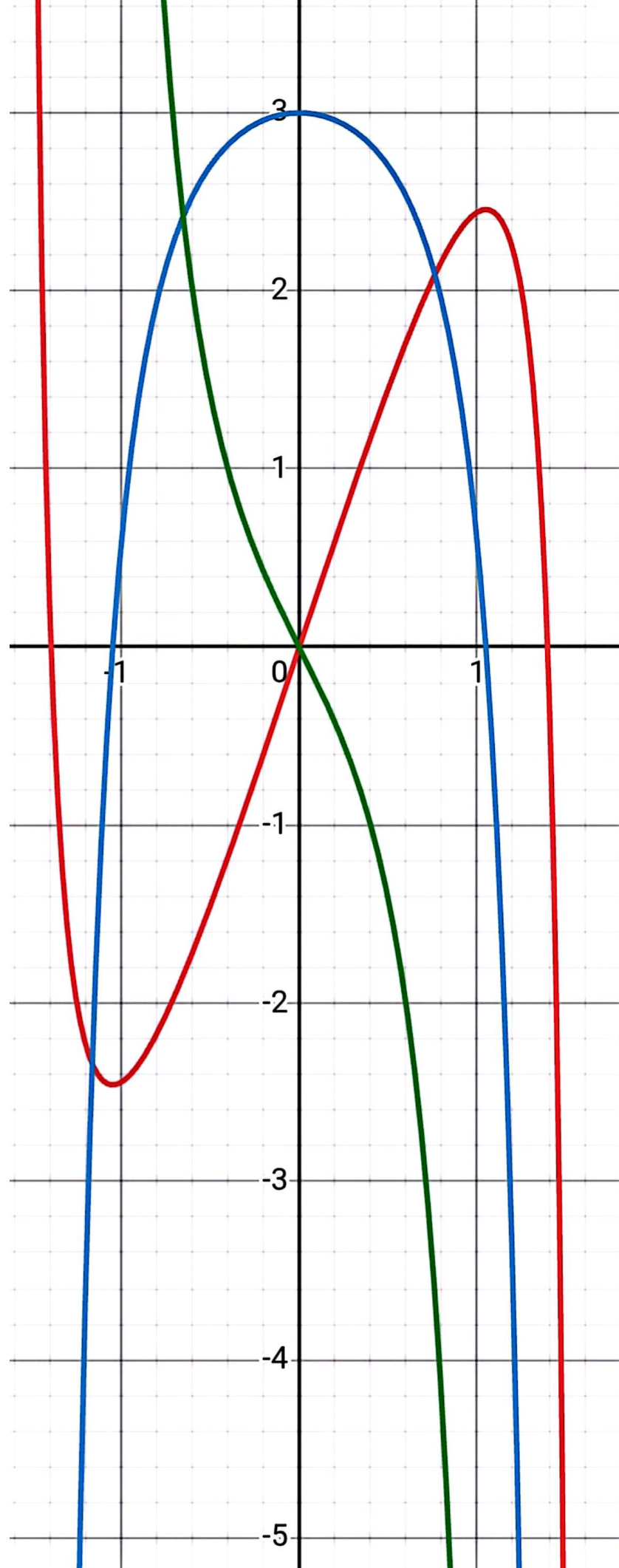
Again Differentiating eqn (ii) w.r.t. x we get,

$$\frac{dy'}{dx} = \frac{d(4)}{dx} - \frac{d\sec^2 x}{d\sec x} \cdot \frac{d\sec x}{dx}$$

$$= 0 + 2\sec x \cdot \sec x \cdot \tan x$$

$$= 2\sec^2 x \tan x$$

b) Check to see that your answers to part (a) are reasonable by comparing the graphs of f , f' and f'' .



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69) At what points on the curve $y = \sin n + \tan n$, $0 \leq n \leq 2\pi$ is the tangent line horizontal?

Soln:-

The tangent line is horizontal when slope is 0. (i.e. $y' = 0$)

$$y = \sin n + \tan n$$

$$\frac{dy}{dn} = \cos n - \sin n$$

then,

Solving for n in interval $(0, 2\pi)$:

$$\tan n = 1$$

Solutions are in quadrant I and III

$$n = \frac{\pi}{4} \text{ and } n = \frac{5\pi}{4}$$

Finding y -coordinates,

$$\text{For } n = \frac{\pi}{4}$$

$$y = \sin\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\text{For } n = \frac{5\pi}{4}$$

$$y = \sin\left(\frac{5\pi}{4}\right) + \tan\left(\frac{5\pi}{4}\right) = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$$

$\therefore$ The points are $\left(\frac{\pi}{4}, \sqrt{2}\right)$ and $\left(\frac{5\pi}{4}, -\sqrt{2}\right)$

70) Find the points on the ellipse $x^2 + 2y^2 = 1$ where the tangent line has slope 1.

Soln:-

$$x^2 + 2y^2 = 1 \quad \text{--- (i)}$$

Differentiating eqn (i) by implicit differentiation we get

$$\frac{d}{dx}(x^2 + 2y^2) = \frac{d}{dx}(1)$$

$$2x + 4y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{4y} = -\frac{x}{2y}$$

When slope is 1

$$1 = -\frac{x}{2y}$$

$$x = -2y$$

Now, substituting $x = -2y$ into eqn (i)

we get,

$$(-2y)^2 + 2y^2 = 1$$

$$4y^2 + 2y^2 = 1$$

$$y^2 = \frac{1}{6}$$

$$y = \pm \frac{1}{\sqrt{6}}$$

Put y in x

$$x = -\frac{2}{\sqrt{6}}$$

$$x = \pm \frac{2}{\sqrt{6}}$$

$$\text{and } y = -\frac{1}{\sqrt{8}}$$

$$x = -2 \left(-\frac{1}{\sqrt{8}} \right)$$

$$x = \frac{2}{\sqrt{8}}$$

$\therefore$ The points on the ellipse $x^2 + 2y^2 = 1$ is $\left(\frac{2}{\sqrt{8}}, \frac{1}{\sqrt{8}} \right)$ and $\left(-\frac{2}{\sqrt{8}}, \frac{1}{\sqrt{8}} \right)$